

Insider Trading Detection on Polymarket: Identifying Informed Betting in Geopolitical Prediction Markets

Rodrigo Arguello

(ra2646)

Shwetanshu Raj

(sr8250)

Course Section: Big Data CS-GY 6513 D

Semester Term: Spring 2026

Prof. Amit Patel · NYU Tandon School of Engineering

Table of Contents

Abstract	3
1. Problem Statement and Objectives	3
1.1 Markets We Focused On	4
1.2 Our Objectives	4
2. Tech Stack We Used	5
2.1 Changes in Technology from Original Proposal	5
3. Methodology and Technical Implementation	5
3.1 Phase 1 - Data Collection	5
3.2 Phase 2 - Data Ingestion and Storage	6
3.3 Phase 3 - Spike Detection	7
3.4 Phase 4 - Bet Tracing	7
3.5 Phase 5 - Prescience Scoring	8
3.6 Phase 6 - Classification and Clustering	8
4. Key Insights and Findings	9
4.1 Validation: rediscovering the publicly charged DOJ case	9
4.2 Pipeline-flagged candidate: Vivaldi007	10
4.3 Pipeline-flagged candidate: 0x1caa6a	11
4.4 Market-maker false positive: 0x4bfb...b8982e	11
4.5 Spike pattern: US Strikes Iran (anticipatory accumulation)	12
4.6 Spike pattern: US Forces Iran (most-flagged active market)	12
4.7 Spike pattern: Maduro Out (cleanest single-shock)	13
4.8 Spike pattern: China Invades Taiwan (stable, no signal yet)	13
4.9 When suspects first trade relative to the spike	14
4.10 Feature correlation heatmap	14
5. Challenges We Faced	15
6. Limitations	15
7. Future Scope	16
8. Lessons Learned	16
9. Appendix	16
9.1 Source Code	16
9.2 Code Execution Instructions	16
9.3 Data Source	17
10. References	17

*** NOTE: Per the rubric, the major insights we could not cover during the presentation (due to the 15-minute limit) are collected under Section 4 - Key Insights and Findings, which also contains the project's primary results: the validation and candidate-wallet findings.*

Abstract

We built a 6 phase batch pipeline to detect insider trading on Polymarket's geopolitical prediction markets, using only on-chain trade data and public news. Trade level data came from Polymarket's CLOB endpoint and the Goldsky on-chain subgraph (333,363 fills across six contracts). The pipeline flags anomalous price movements with a rolling z-score over 5-minute volume-weighted average price (VWAP) buckets in PySpark, traces the wallets responsible for large directional bets in the 24-hour pre-spike window, scores those wallets against a GDELT news timeline queried via BigQuery, and classifies the resulting 11,373 wallet feature vectors with a Random Forest in Spark MLlib.

3 of the 6 markets resolved during our collection window, which gave us real ground truth to validate against. Our central finding has three parts. First, the pipeline surfaces a wallet (0x31a56e9E61eD21De0..., the soldier's wallet) whose public Polymarket profile is consistent with the soldier named in the April 2026 DOJ indictment: account created December 2025, four predictions, +\$409,882 lifetime profit, single-market exposure on Maduro-Out. Second, the same pipeline surfaces two additional unilateral candidates - Vivaldi007 (0x5ed3...b19e, RF probability 0.942) and 0x1caa6a...46a0 (RF probability 0.983) - whose pre-spike fingerprints match the soldier's case and whose lifetime profit-and-loss confirms sustained directional success. Third, the highest-volume address on the platform (0x4bfb...b8982e) is correctly downranked by the classifier to 0.024 once behavioral features are available, because it is a market-making contract that profits from the spread regardless of outcome.

Keywords: *prediction markets, insider trading detection, PySpark, BigQuery, GDELT, Spark MLlib, z-score spike detection, wallet attribution, K-Means clustering, Random Forest classification*

1. Problem Statement and Objectives

Polymarket processed over \$9 billion in volume during the 2024 U.S. election cycle. Users buy shares in real-world outcomes and the price reflects a crowd-sourced probability. All trading happens on the Polygon blockchain where every wallet is pseudonymous: there is no identity verification, no know-your-customer requirement, and no regulatory oversight comparable to traditional finance.

That structure invites insider trading. Someone with advance knowledge of a geopolitical event (a military strike, a leadership change, a diplomatic agreement) can bet heavily on the right side of a contract before the news is public. Because every trade is on-chain and timestamped, the pattern is detectable from public data alone, even without knowing who placed it.

Example: in April 2026 the U.S. Department of Justice charged Army Special Forces Master Sergeant Gannon Van Dyke for betting \$33,000 on Maduro's capture days before Operation Absolute Resolve, profiting approximately \$409,000. It is the first DOJ prosecution of insider trading on a prediction market.

POLITICS

U.S. soldier arrested for Polymarket bets on Maduro capture that won \$400K, Justice Department says

PUBLISHED THU, APR 23 2026 6:01 PM EDT | UPDATED FRI, APR 24 2026 8:37 AM EDT

Dan Mangan
@DANMANGAN/CNBC
@DANMANGAN

Jim Forkin

SHARE

KEY POINTS

- A U.S. Army Special Forces master sergeant was arrested on an indictment accusing him of using classified information to make bets that won him \$400,000 on the Polymarket prediction market related to the U.S. military mission that captured Venezuelan leader Nicolás Maduro, the Department of Justice said.
- Gannon Ken Van Dyke "was involved in the planning and execution of Operation Absolute Resolve," which apprehended Maduro and his wife, Cilia Flores, in early January, the DOJ said.
- The Commodity Futures Trading Commission separately charged Van Dyke in a civil complaint for allegedly using classified nonpublic information to make the wagers related to the Maduro capture mission.

TRENDING NOW



Meme stock GameStop makes \$56 billion offer for eBay in bid to rival Amazon



Rudy Giuliani, former New York mayor and Trump lawyer, hospitalized in critical condition



Trump's Hormuz gambit lands with a whimper

CNBC, April 23, 2026

We focused on geopolitical markets because the information asymmetry is unusually legible: each contract resolves on a discrete, timestamped event, and the gap between when an insider knows the outcome and when the public does is directly measurable.

1.1 Markets We Focused On

Market	Description	Volume	Status
US Strikes Iran by Feb 28	Resolves YES if the US initiates a drone, missile, or air strike on Iranian soil or any official Iranian embassy/consulate by Feb 28, 2026.	\$89.7M	Resolved YES
Maduro Out by Jan 31	Resolves YES if Venezuela's president, Nicolas Maduro, is removed from power between Dec 11 2025 and Jan 31 2026.	\$56.6M	Resolved YES
Khamenei Out by Feb 28	Resolves YES if Iran's Supreme Leader, Ali Khamenei, is removed from power for any length of time within the market's timeframe.	\$131.1M	Resolved YES
US Forces Enter Iran by Mar 31	Resolves YES if active US military personnel physically enter Iran by the listed date.	\$11.0M	Active
US-Iran Ceasefire by Jun 30	Resolves YES if there is a publicly announced bilateral ceasefire between the United	\$1.1M	Active

	States and Iran by Jun 30, 2026.		
China Invades Taiwan by 2026	Resolves YES if China commences a military offensive on any portion of Taiwan by December 31, 2026.	\$11.0M	Active

Markets that we focused on. Three resolved during our collection window, providing ground truth for validation; three remain active, providing forward-test candidates.

1.2 Our Objectives

1. Collect tick-by-tick trade data across six geopolitical Polymarket contracts (~\$300M in combined volume).
2. Flag anomalous price spikes using rolling z-score analysis over 5-minute VWAP buckets in PySpark.
3. Trace the wallets behind large directional bets in the 24 hours before each spike.
4. Score those wallets against GDELT's public news timeline and check how consistently they moved before news broke.
5. Train a Random Forest classifier (Spark MLlib) to assign insider probability scores across all 11,373 suspect wallets.
6. Cluster wallets by behavior to identify archetypes (serial insiders, high-stakes one-offs, opportunists).
7. Validate the pipeline against the publicly charged DOJ case on the Maduro Out market.

2. Tech Stack We Used

Technology	Purpose
PySpark	Distributed DataFrame processing for 333K+ row ingestion, VWAP aggregation, and window-function-based spike detection.
Google BigQuery	Cloud data warehouse for all structured tables. Native host of GDELT for news timeline queries.
Google Cloud Storage	Raw JSONL file backup.
Spark MLlib	Random Forest classifier for insider probability scoring and K-Means clustering for behavioral segmentation.
GDELT via BigQuery	Global event news timeline queried directly in BigQuery.
Matplotlib	Visualisation.

2.1 Changes in Technology from Original Proposal

Our original proposal listed HDFS, Hive, and Apache Tez, which we dropped in favor of a PySpark + BigQuery architecture for two reasons:

- BigQuery made the Hive/Tez layer redundant. Hive was proposed as the warehouse for cross-dataset joins. BigQuery is a fully managed columnar warehouse that handles those same joins with standard

SQL. Given that GDELT is natively hosted on BigQuery, querying it alongside our own tables in the same environment eliminated a data-transfer step and simplified the pipeline.

- The NYU Dataproc cluster did not allow both team members to access the same environment simultaneously. Running HDFS and Hive jobs would have required one of us to be sole operator. Migrating to BigQuery gave us both equal authenticated access to the same tables from our local machines at all times.

3. Methodology and Technical Implementation

We implemented this project in 6 phases.

3.1 Phase 1 - Data Collection

Goal: collect all raw trade and market data from Polymarket APIs and the Goldsky on-chain subgraph. Six geopolitical markets were selected.

Collection scripts (in collect/):

- `fetch_metadata.py`: pulls contract metadata (title, description, resolution outcome) from the Gamma API.
- `fetch_prices.py`: pulls hourly price history from the CLOB API.
- `fetch_wallets.py`: pulls all on-chain fills with maker/taker addresses from the Goldsky GraphQL subgraph. This is the primary source for wallet attribution because the standard Polymarket APIs do not expose wallet addresses on public trades.
- `extract_ticks.py`: derives a per-fill tick dataset, computing $\text{implied_price} = \text{takerAmountFilled} / \text{makerAmountFilled}$ and normalizing micro-USDC amounts to USD ($\div 1,000,000$), writing one tick record per fill.

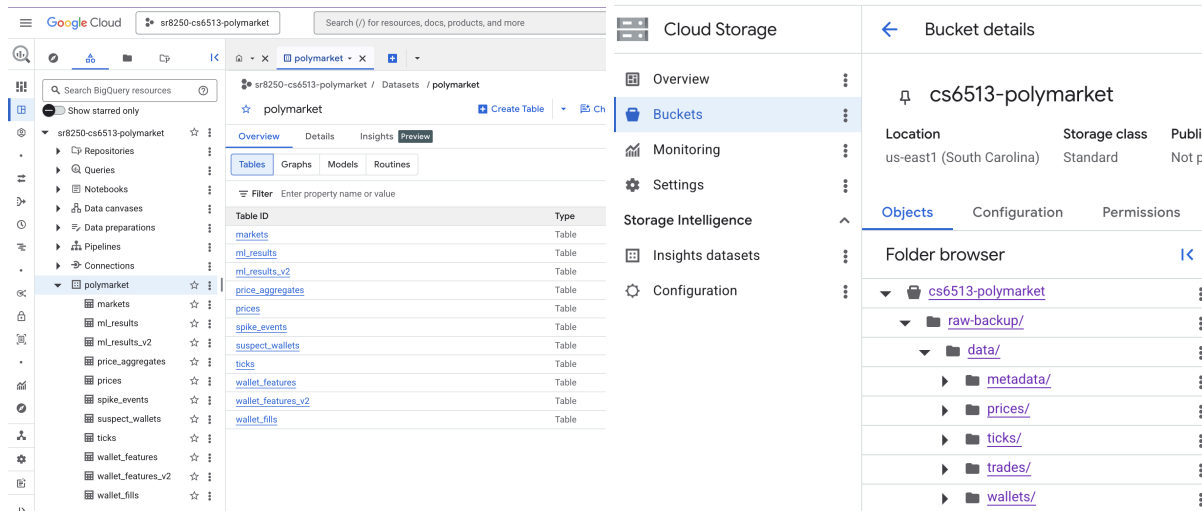
Output: ~333,363 raw fills written to JSONL files in `data/wallets/` and `data/ticks/` (~666K data points total).

3.2 Phase 2 - Data Ingestion and Storage

Goal: load all JSONL data into BigQuery using PySpark via the script `pipeline/ingest.py`.

- Backed up all JSONL files to a GCS bucket using the GCS Python client.
- Loaded metadata, wallet fills, ticks, and price data into BigQuery tables using PySpark with the BigQuery connector. Cleaning was limited to type-casting Unix epoch strings to `TIMESTAMP` and dropping fills with unparseable amounts.

Result: 5 tables (`markets`, `prices`, `ticks`, `wallet_fills`, `spike_events`) and raw data backup on gcp bucket



3.3 Phase 3 - Spike Detection

Goal: detect anomalous price movements that preceded known resolution events. Notebook: analysis/detect.ipynb.

- Read polymarket.ticks from BigQuery via Spark connector (329,373 rows).
- Aggregated ticks into 5-minute VWAP buckets per (market_slug, side) using F.window(), producing 65,985 buckets.
- Computed a rolling baseline over the preceding 20 buckets: rolling_mean and rolling_std using Spark SQL window functions (rowsBetween(-20, -1)).
- Computed $z_score = (vwap - rolling_mean) / rolling_std$.
- Flagged spikes where $z_score > 3.0$, volume_usd $\geq$ \$100, and z-score is non-null.

Output: spike data per market saved to spike_events table on BigQuery.

3.4 Phase 4 - Bet Tracing

Goal: for each spike, identify wallets that traded on the correct side in the preceding 24 hours. Notebook: analysis/trace.ipynb.

- Loaded polymarket.spike_events and polymarket.ticks from BigQuery.
- For each spike, defined a 24-hour look-back window = [spike_timestamp - 24h, spike_timestamp).
- Performed a temporal join to find all ticks in the look-back window matching the spike's market and direction.
- Takers are wallets on the spike side (e.g. bought YES before a YES spike). Makers are wallets on the opposite side (sold NO before a YES spike), which is also a directional bet.
- Aggregated per (wallet, spike): total USD traded, fill count, average price, earliest trade time.
- Calculated hours_before_spike to measure how early the wallet started accumulating.
- Filtered out wallets below a minimum USD threshold (\$100).

Result: 13,068 distinct suspect wallet addresses across 128,096 wallet-spike pairs (after filtering). The single highest-volume address in this set is 0x4bfb41d5b3570defd03c39a9a4d8de6bd8b8982e, which Section 4.4 examines in detail.

3.5 Phase 5 - Prescience Scoring

Goal: measure how consistently each wallet traded before public news on the same topic. Notebook: analysis/score_v2.ipynb.

- Loaded polymarket.suspect_wallets.
- Queried GDELT eventmentions (gdelt-bq.gdeltv2.eventmentions joined with gdelt-bq.gdeltv2.events) per market using keyword mappings, returning article-level timestamps with 15-minute precision via MentionTimeDate.
- For each suspect wallet entry, performed a temporal join to find the nearest GDELT mention that appeared after their first_bet_timestamp on the same market topic.
- Computed time_before_news_hours and bet_before_news flag.

Aggregated a per-wallet feature vector:

- prescience_score: fraction of bets placed before any relevant news mention.
- win_rate_v2: dollar-weighted fraction of capital deployed on the winning side, computed only on resolved markets.
- avg_hours_before_news: how far in advance of news suspects typically bet.
- avg_bet_size_usd, num_spikes_flagged, num_markets_traded, taker_fraction.
- avg_entry_price, avg_hours_before_spike.

Result:

- 11,373 total scored wallets.
- 96.4% have prescience_score = 1.0. This is not a victory for the metric: GDELT coverage is dense for these topics, and most bets will find a subsequent matching article within a few hours. We treat the ceiling as the signal: a wallet that has prescience_score = 1.0 AND a high win_rate_v2 AND a large avg_hours_before_news is meaningfully different from a wallet with prescience_score = 1.0 alone.
- 102 confirmed reactors (prescience_score < 0.3) - wallets that consistently bet after news appeared.
- The 0x4bfb... wallet (Section 4.4) had prescience_score = 0.991, win_rate_v2 = 0.42, avg_hours_before_news = 1.9h - high prescience but a near-50% directional win rate, which is the fingerprint of a market-making contract rather than an insider.

3.6 Phase 6 - Classification and Clustering

Goal: assign insider probability scores and identify behavioral clusters. Notebook: analysis/classify.ipynb.

- Labeled training data using a heuristic rule: prescience_score > 0.7 AND win_rate_v2 > 0.65 -> label 1. Wallets in our curated known_suspects.jsonl (populated from public reporting) were also confirmed label 1. All others -> label 0.
- Split labeled data 70/30 and trained a Random Forest (Spark MLlib, 100 trees) to predict the label from the 9 wallet features.
- The trained model outputs a label_probability (0 to 1) for every wallet.

- Ran K-Means (K=4) on label-1 wallets only to identify behavioral archetypes; three of the four clusters received clean human-readable interpretations.

Random Forest performance on the held-out 30% test split:

Metric	Value
Accuracy	0.9903
Precision	0.9906
Recall	0.9903
F1	0.9904
AUC-ROC	0.9995

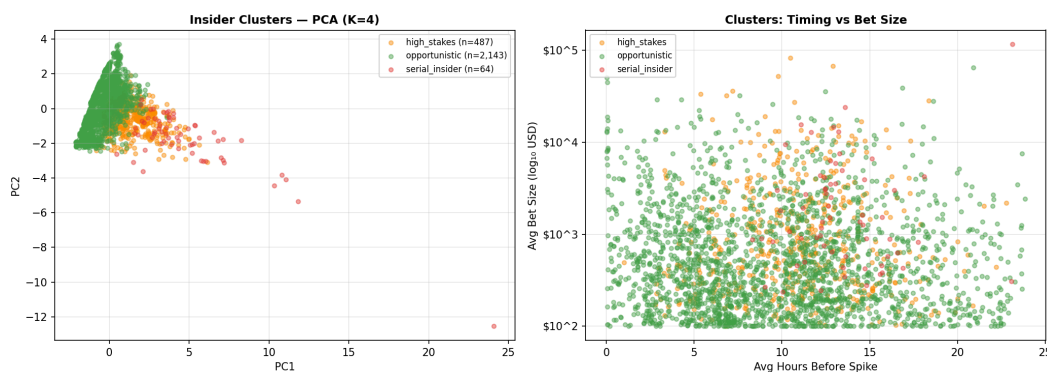
Confusion matrix (label vs prediction):

	Predicted 0	Predicted 1
True 0	2,481	30
True 1	2	802

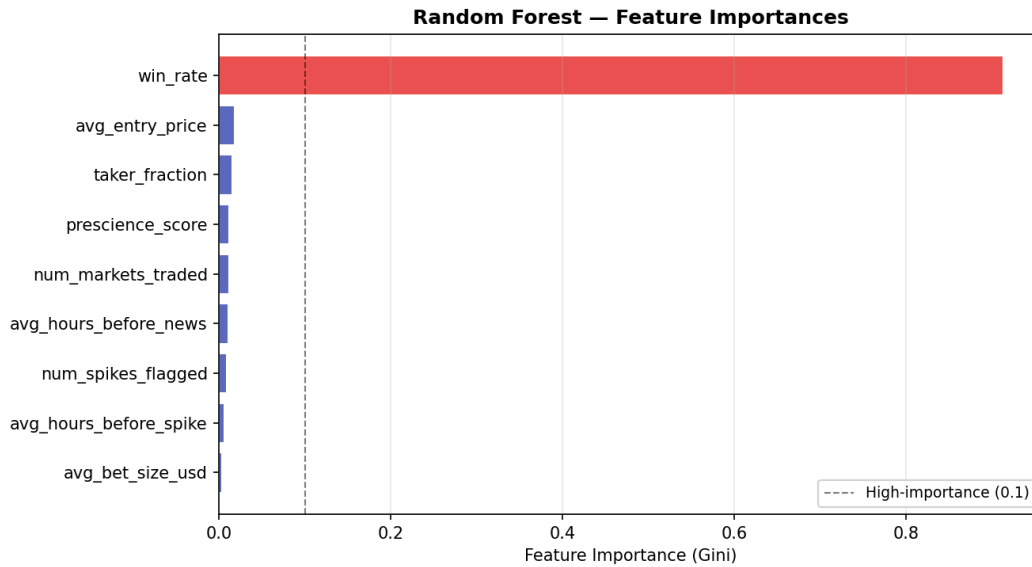
Caveat: the training labels are produced by the rule `prescience_score > 0.7 AND win_rate_v2 > 0.65`, and two of the nine classifier features are precisely these two quantities. The Random Forest's near-perfect metrics partly reflect the fact that it can recover the labeling rule from its inputs. The more meaningful evidence that the classifier carries information is the case-by-case behavior in Section 4: it correctly downranks the highest-volume address (Section 4.4) and assigns the dataset's highest scores to wallets whose pre-spike fingerprints match the soldier's case (Sections 4.2-4.3).

K-Means clusters on the label-1 subset (three clean interpretations):

- `serial_insider` (n = 64): multi-market, high stakes, consistent.
- `high_stakes` (n = 487): single-market large directional bets.
- `opportunistic` (n ~ 2,143): single-market, lower capital.



K-Means clusters on the label-1 subset projected to two principal components. Serial insiders separate cleanly from opportunistic and high-stakes wallets along PC1.



Random Forest feature importances. Win rate and prescience score dominate as expected, but avg_hours_before_news and num_markets_traded carry meaningful weight.

4. Key Insights and Findings

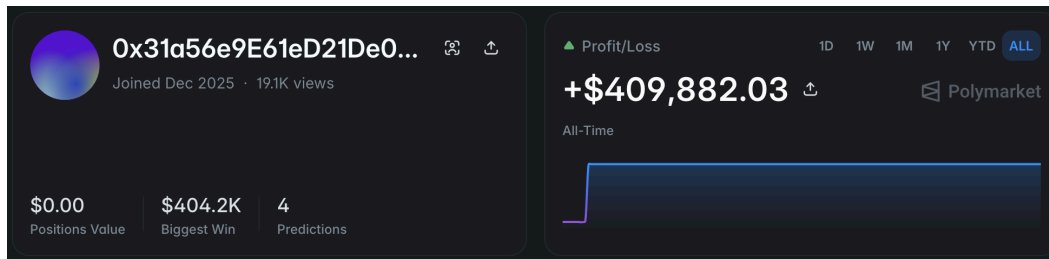
Sections 4.1-4.4 cover the wallet-level findings: the validation against the publicly charged DOJ case, two unilateral candidates the pipeline surfaced, and one negative example (a market-making contract correctly downranked). Sections 4.5-4.10 cover market-level patterns and cross-cutting analyses.

4.1 Validation: rediscovering the publicly charged DOJ case

The Maduro Out market is the contract named in the April 2026 DOJ indictment. Our spike detector flags the resolution-day price jump at $z = 33.3$ (price moves from 12c to 63c in five minutes on Jan 3, 2026 at 01:15 UTC). The bet-tracer's 24-hour pre-spike window produces a suspect pool of 843 wallets for this spike.

Within that pool, the wallet 0x31a56e9E61eD21De0... (the soldier's wallet) appears with 39 fills totaling roughly \$87K, all on the YES side, and traces the entire 22-hour pre-resolution accumulation curve. Its public Polymarket profile is consistent with the description in the DOJ indictment on every observable dimension:

- Account creation date: December 2025 (joined shortly before Operation Absolute Resolve).
- Lifetime profit: +\$409,882 (the indictment described approximately \$400,000 in profit).
- Number of predictions: exactly 4 (consistent with a small, focused unilateral bettor).
- Single-market exposure on Maduro Out (consistent with the DOJ description of a single bet).
- Biggest single win: \$404.2K (matches the resolution-day payoff).



Polymarket profile for 0x31a56e9E61eD21De0... (the soldier's wallet). Joined Dec 2025, 4 predictions, biggest win \$404.2K, lifetime P&L +\$409,882.

The DOJ filing did not name the wallet directly, so we cannot claim definitive identification, but the pipeline's output and the public profile align on every observable dimension. This wallet would not have surfaced from raw volume alone (it traded only \$87K, far below the platform's high-volume addresses); it surfaces because its behavioral fingerprint - single-market unilateral exposure entered hours before a confirmed information shock - is exactly what the pipeline was designed to catch.

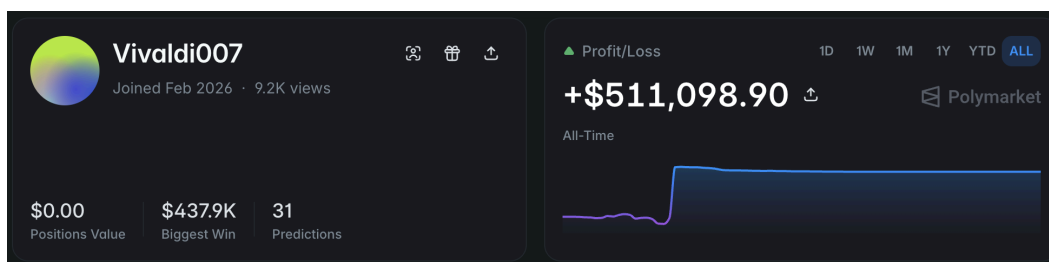
Three additional public cases on these markets exist but were not the focus of our validation: the user 'Magamyman' on Khamenei Out (NPR-reported, \$553K profit), a cluster of six fresh wallets on US Strikes Iran identified by Bubblemaps on-chain analysis (~\$1.2M combined), and two Israeli defendants charged by Israeli prosecutors for trading on US Strikes Iran with classified military intelligence (\$150K+ combined). The soldier case is the only one with a U.S. DOJ indictment, which is why we use it as the primary validation target. The full list is tracked in config/known_suspects.jsonl in the repository.

4.2 Pipeline-flagged candidate: Vivaldi007 (0x5ed3...b19e)

Vivaldi007 received Random Forest probability 0.942 and falls into the serial_insider K-Means cluster. The wallet is an Iran-region specialist that traded four related markets and won all four. On US Strikes Iran the wallet entered at an average of 14c approximately 115 hours before the largest pre-resolution spike, deployed \$4,687 across 37 taker fills purely on the YES side, and recorded its biggest single position at +\$437,930 profit.

Lifetime returns across the four Iran/Israel markets:

- 684% on the first market.
- 571% on the second.
- 1,011% on the third.
- 88% on the fourth (Khamenei Out, \$98K profit on \$111K traded).

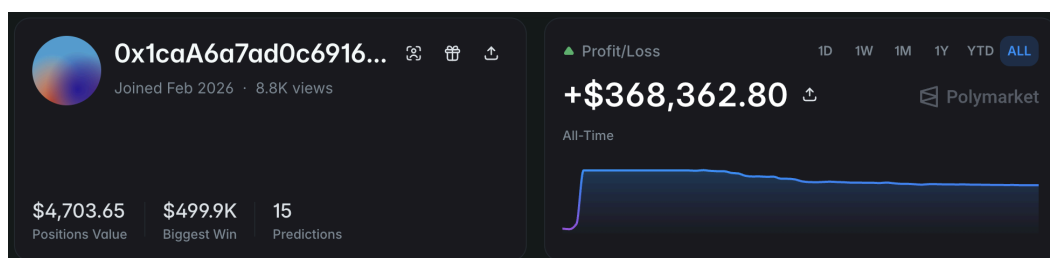


Polymarket profile for Vivaldi007 (0x5ed3...b19e). +\$511,098.90 lifetime profit, \$437.9K biggest single win, 31 predictions across multiple Iran-region markets.

The pre-spike behavioral fingerprint - single-direction taker fills entered well before the spike at low entry prices - mirrors the soldier's pattern, applied to neighboring geopolitical contracts. Vivaldi007 was not previously discussed in public reporting; the pipeline surfaced it from on-chain data alone.

4.3 Pipeline-flagged candidate: 0x1caa6a...46a0

0x1caa6a...46a0 received Random Forest probability 0.983, the highest score in the dataset, and falls into the opportunistic K-Means cluster (high score with single-market exposure). On US Strikes Iran the wallet entered at an average of 10c approximately 37 hours before the largest pre-resolution spike, deployed \$12,711 across 9 taker fills purely on the YES side, and recorded a lifetime profit of +\$368,362.80 with a per-market win rate of 1.0.



Polymarket profile for 0x1caa6a7ad0c6916... +\$368,362.80 lifetime profit, \$499.9K biggest single win, 15 predictions concentrated on US Strikes Iran.

The compact bet, the early entry, the single-direction taker exposure, and the lifetime payoff are the same archetype the soldier exemplifies, applied to a different geopolitical contract. Both Vivaldi007 and 0x1caa6a received label 1 from the heuristic rule and label 1 from the classifier; their pre-spike fingerprints are not exotic outliers, they are the same shape as the public DOJ case.

4.4 Market-maker false positive: 0x4bfb...b8982e

The single highest-volume address in the suspect pool is 0x4bfb41d5b3570defd03c39a9a4d8de6bd8b8982e, which appears as the largest spender in the 24-hour pre-spike window across all three resolved markets. A naive raw-volume ranking would place this address at the top of the suspect list. The Random Forest probability for this wallet is 0.024.

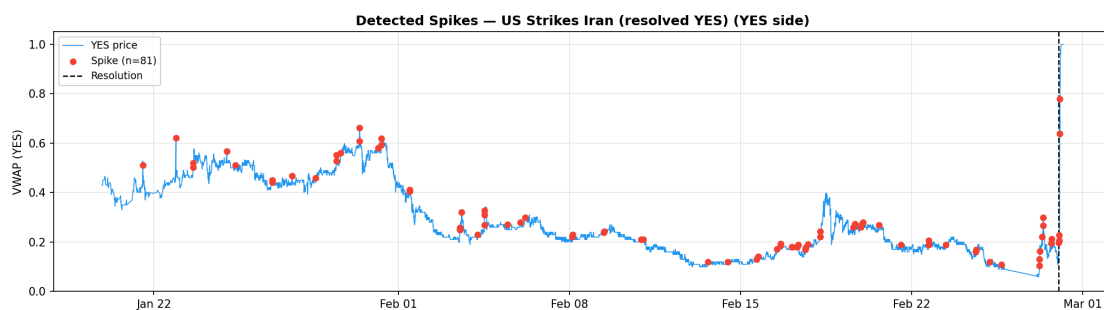
Feature vector for 0x4bfb...b8982e:

Feature	Value
num_spikes_flagged	562
num_markets_traded	6
avg_bet_size_usd	116,158.79
num_fills_total	268,389
avg_entry_price	0.4984
avg_hours_before_spike	23.13
prescience_score	0.9911
avg_hours_before_news	1.9
win_rate_v2	0.42
taker_fraction	1.0

These are the fingerprints of a market-making contract that profits from the bid-ask spread regardless of which side eventually wins: high volume, an average entry price near the mid (0.50), high prescience (it is always trading, so it is always before any news), but a near-50% directional win rate (0.42). An on-chain check via `eth_getCode` confirms the address holds bytecode rather than being an externally owned account.

We include this address explicitly because it is the most informative negative example in the dataset. The classifier's job is not to surface high volume, it is to surface high directional informedness. The fact that the classifier separates these two cases (the soldier and the two pipeline-flagged candidates received the highest scores; this wallet received 0.024) is the strongest evidence that the per-wallet feature vector carries useful structure.

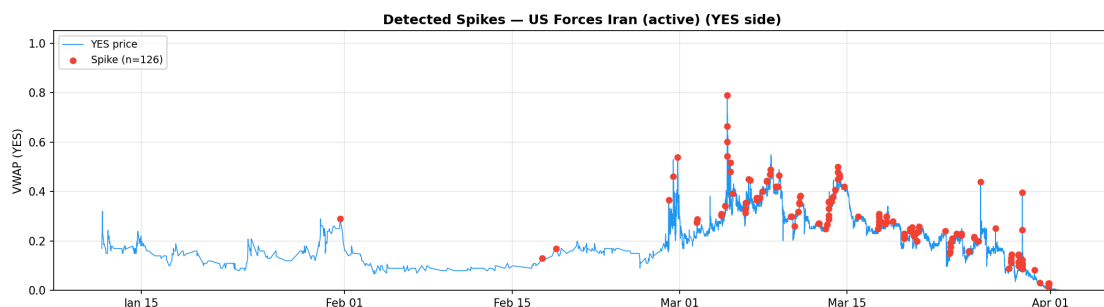
4.5 Spike pattern: US Strikes Iran (anticipatory accumulation)



Detected spikes on US Strikes Iran (YES side). Red dots mark buckets with $z > 3$.

- Red dots scatter in the weeks before resolution with a clear cluster at the Feb 28 resolution day.
- An earlier spike on Jan 22 reaches $z = 17.89$, 36 days before resolution. Within the resolved-market subset this is the strongest anticipatory signal in our data, and it is the spike where Vivaldi007 and 0x1caa6a both flagged.
- The pre-resolution spikes suggest informed buyers were accumulating YES contracts weeks before the strike.

4.6 Spike pattern: US Forces Iran (most-flagged active market)



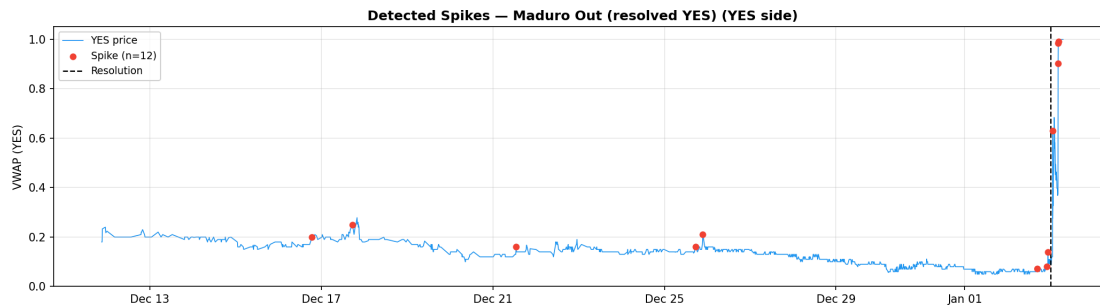
Detected spikes on US Forces Iran (YES side). Most flagged market in the dataset ($n = 126$).

- Most red dots of any market, spread across the timeline.
- The single largest spike in the entire dataset ($z = 60.9$) lives here, on March 26 — price moves from 0.19 to 0.44 in one 5-minute window on \$8,521 of volume. That is nearly 2x the magnitude of the

Maduro resolution-day shock and roughly 5x the US Strikes Iran resolution spike, in a market that has not yet resolved.

- Multiple smaller spikes in late March 2026 surround the largest.
- This market showed abnormal activity when the US-Iran situation was actively escalating, providing a forward-test candidate for the classifier once the market resolves.

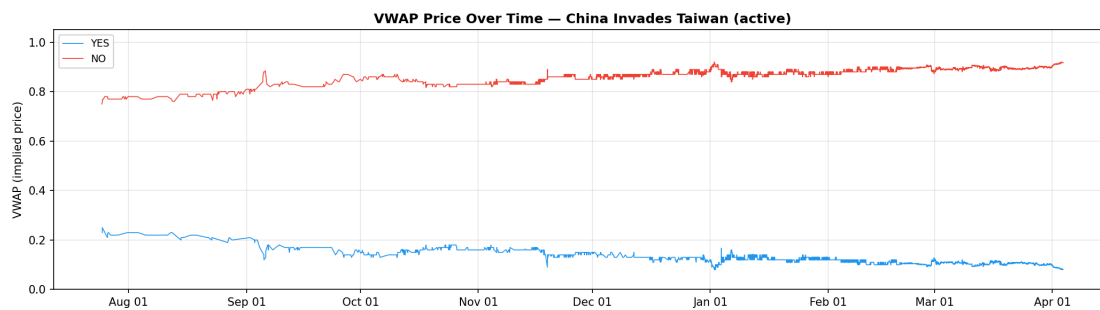
4.7 Spike pattern: Maduro Out (cleanest single-shock)



Detected spikes on Maduro Out. A single dominant cluster at the Jan 3 resolution moment.

- The largest pre-resolution spike in the resolved-market subset ($z = 33.3$, price 0.12 to 0.63) lands precisely on Jan 3, 2026 at 01:15 UTC, on \$17,930 of traded volume.
- Three hours later, a second wave at 04:20 UTC moved the price 0.375 to 0.902 with \$152,184 in volume but only $z = 10.35$. The informed money was small and early; the chasers showed up late with eight times the dollar volume but a much weaker statistical signature. The z -score separates them; raw volume alone would not.
- An earlier spike on Jan 2 at 22:40 (price 0.08 to 0.14) sits approximately 2.5 hours before the Jan 3 window opens - suspicious pre-resolution buying that the trace step picks up.
- This is the cleanest information-shock signature in our dataset, and the spike that surfaces the soldier's wallet (Section 4.1).

4.8 Spike pattern: China Invades Taiwan (stable, no signal yet)

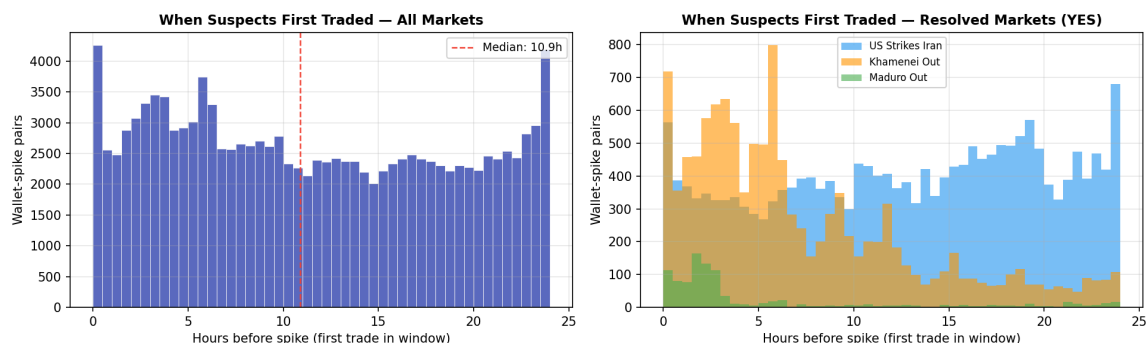


VWAP price over time: YES (blue) stays low and stable; NO (red) is correspondingly high.

- The market does not expect any invasion: YES price is low and stable, NO is correspondingly high.
- Any sharp YES spike here would be a strong signal for insider trading if the market subsequently resolves YES.

- There are very few spikes detected, and they are scattered with no obvious clustering around a specific event yet. This market is a long-running forward-test candidate.

4.9 When suspects first trade relative to the spike



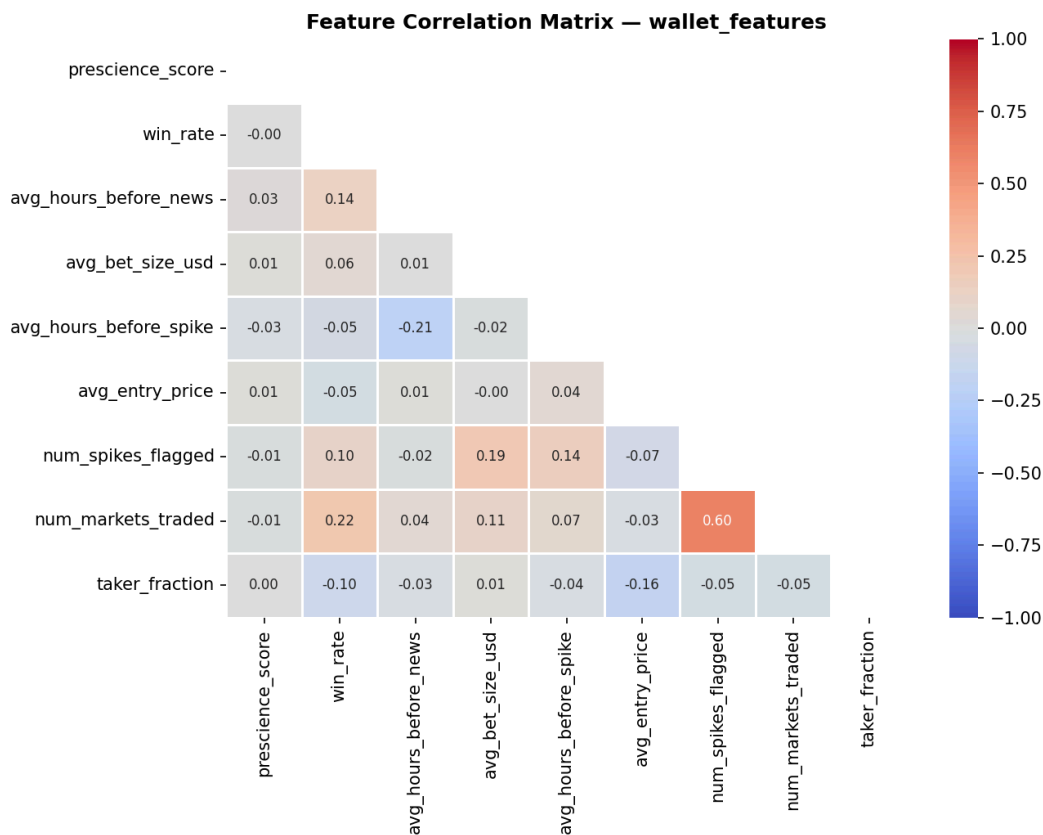
Distribution of hours before spike at first qualifying trade, aggregated across all suspect wallet-spike pairs.

The aggregated distribution has two peaks at opposite ends of the 24-hour window:

- Peak at 0-0.5h (~4,200 pairs): wallets that placed their first qualifying trade within the last 30 minutes before the spike. These are reactive traders, likely responding to breaking news or a price move that was already starting.
- Middle section (1-22h): fairly flat at 2,200-3,700 pairs per 30-minute bin, with a secondary rise around 3-6h. The roughly flat shape across 10+ hours means suspect activity is genuinely spread across the day, not concentrated at one phase.
- Peak at 23.5-24h (~4,000 pairs): any wallet that was accumulating for longer than 24 hours gets its first qualifying trade clipped to the start of the look-back window. These wallets did not necessarily start exactly 24h before; they were simply already active when the window opened.
- Median: 10.9h. The typical suspect wallet first traded about 11 hours before its spike, roughly halfway through the window.

Per-market breakdown: Khamenei suspects cluster at 0-1h (reactive, dropping off after 8h), while US Strikes Iran has substantial counts at 15-24h (those wallets were positioning well in advance, the more anticipatory pattern). Khamenei looks reactive; US Strikes Iran looks anticipatory.

4.10 Feature correlation heatmap



Feature correlation matrix (wallet_features). Most pairs are uncorrelated, indicating the 9 features carry largely independent information.

- Nearly all correlations are close to 0, meaning the 9 features are largely independent and carry unique info.
- num_spikes_flagged vs num_markets_traded = 0.60: wallets active in more markets naturally appear before more spikes.
- win_rate vs num_markets_traded = 0.22: wallets trading more markets have slightly better win rates, possibly because diversification across markets correlates with better information.

5. Challenges We Faced

- The standard Polymarket CLOB and Data APIs expose trade records but not wallet addresses. To get maker/taker attribution (the foundation of the entire bet-tracing phase), we had to discover and use the Goldsky GraphQL subgraph, which indexes on-chain fills and includes wallet addresses. This required pagination handling (GraphQL cursor-based pagination), rate-limit management, and a custom matching step to align subgraph records with CLOB tick records by transaction hash.
- Deciding on the win-rate formula: count-based vs dollar-weighted. Initially the win rate counted how many times a wallet appeared in YES spike windows versus total spike windows in resolved markets. The problem: the spike detector flags both YES and NO spikes on a volatile resolution day. For Khamenei on Feb 28 it found 28 YES spikes and 28 NO spikes, giving roughly equal counts on each

side. The formula returned a win rate of ~ 0.5 for any continuously active wallet, regardless of whether all their actual money was on the correct side. Fix: weight by dollars deployed instead of appearance counts. $\text{win_rate_v2} = (\text{total USD on YES side in resolved markets}) / (\text{total USD in all resolved market windows})$.

- The first GDELT query used the events table, which only records SQLDATE (no time). For every bet placed on a given day, the next relevant news event would always be found on the same or next calendar day, making the prescience score meaningless. Fix: switched to the eventmentions table, which records MentionTimeDate at 15-minute crawl precision (format: YYYYMMDDHHMMSS).

6. Limitations

- The pipeline runs on six markets selected for resolution clarity and event diversity, not on the full Polymarket universe. The classifier's behavior on a thousand markets remains untested.
- The directional win-rate-driven classifier cannot detect bilateral wallets that profit from the bid-ask spread regardless of which side eventually wins, as the market-maker example in Section 4.4 demonstrates. A net-directional-exposure metric would more meaningfully address the bilateral blind spot.
- GDELT keyword matching is brittle for novel events (a coup with no prior coverage produces no GDELT mentions, and the prescience score becomes mechanically high for any bet on it).
- No sybil clustering: a single actor running multiple wallets appears as multiple independent suspects, inflating the suspect pool and diluting individual classifier scores.
- Label-rule circularity: the training labels are produced by the rule $\text{prescience_score} > 0.7$ AND $\text{win_rate_v2} > 0.65$, and two of the nine classifier features are precisely these two quantities. The Random Forest's near-perfect test metrics partly reflect the fact that it can recover the labeling rule from its inputs.

7. Future Scope

- Replace the directional win rate with a net-directional-exposure metric that integrates signed position over time. This would address the bilateral blind spot illustrated by the 0x4bfb... market-maker contract, which the current classifier correctly downranks but which future bilateral insiders could potentially hide behind.
- Wallet clustering by on-chain funding source. Multiple addresses funded from the same source may represent the same actor; cross-wallet correlation would tighten the suspect pool and merge sybil identities before scoring.
- Forward-validate the classifier on the three active markets (US Forces Iran, US-Iran Ceasefire, China Invades Taiwan) once they resolve, using the current classifier outputs as pre-registered predictions.
- Port the spike detector and bet tracer to Spark Structured Streaming so the same logic runs in near-real-time micro-batches against the Polymarket API rather than after the fact.
- Expanding to all geopolitical contracts on Polymarket (100+ at any given time) would provide more training examples for the classifier.

8. Lessons Learned

- API design determines what questions we can ask. Switching to the Goldsky subgraph instead of the standard Polymarket API unlocked the entire bet-tracing phase; the default API has no wallet-level attribution, so we could detect a price spike but could not connect it to anyone.
- Test features against a known case before using them downstream. The count-based win rate formula looked correct initially but was wrong for any wallet active across multiple spike windows. We only caught it in Phase 6 when the classifier gave the most suspicious wallet in the dataset (the soldier's wallet pattern) a near-zero insider score. Checking the formula against the prime suspect in Phase 5 would have saved a full re-run of two phases.
- Raw volume is not the signal. The highest-volume address in the suspect pool is a market-making contract; the actual high-information wallets traded much smaller amounts. Building features that capture directional informedness rather than activity volume is what made the classifier useful.

9. Appendix

9.1 Source Code

The full source code, with run instructions and the canonical README, is available at:

<https://github.com/RArgu/CS-GY6513-FinalProject>

The README in the repository root contains the run instructions and the high level code logic walkthrough; the steps below mirror that README and reproduce the pipeline end-to-end on a fresh machine.

9.2 Code Execution Instructions

```
# 1. Environment
python3 -m venv venv
source venv/bin/activate
pip install pyspark google-cloud-bigquery google-cloud-storage \
    db-dtypes pandas matplotlib pyarrow ipykernel

# 2. GCP authentication
gcloud auth application-default login

# 3. BigQuery Spark connector JAR
mkdir -p jars
cd jars
curl -O https://storage.googleapis.com/spark-lib/bigquery/\
    spark-bigquery-with-dependencies_2.12-0.36.1.jar
cd ..

# 4. Phase 1 - Data collection
python3 collect/fetch_metadata.py
python3 collect/fetch_prices.py
python3 collect/fetch_wallets.py
python3 collect/extract_ticks.py

# 5. Phase 2 - ingest to BigQuery
```

```
python3 pipeline/ingest.py
```

```
# 6. Phases 3-6 - analysis notebooks (run in order)
#   analysis/detect.ipynb      spike detection
#   analysis/trace.ipynb      bet tracing
#   analysis/score_v2.ipynb    prescience scoring
#   analysis/classify.ipynb    classification + clustering
```

9.3 Data Source

The data can be regenerated by running the scripts under the collect/ folder. If you want the exact dataset we used, the collected JSONL bundle (~48 MB) is hosted at:

https://drive.google.com/file/d/1YNleGjtsZ9gkrAiYkxF4u2ujaBQpPFk_/view?usp=drivesdk

10. References

1. Polymarket: <https://polymarket.com>
2. Polymarket CLOB API Documentation: <https://docs.polymarket.com>
3. Polymarket Gamma API (Contract Metadata): <https://gamma-api.polymarket.com>
4. Polymarket Data API (Trade Records): <https://data-api.polymarket.com>
5. Goldsky Subgraph Documentation: <https://docs.polymarket.com/market-data/subgraph>
6. GDELT Project: <https://www.gdeltproject.org>
7. GDELT 2.0 on BigQuery: <https://blog.gdeltproject.org/gdelt-2-0-our-global-world-in-realtime/>
8. Hanson, R. (2003). Combinatorial Information Market Design. George Mason University.
9. Kyle, A. S. (1985). Continuous Auctions and Insider Trading. *Econometrica*, 53(6), 1315-1335.
10. Wolfers, J., & Zitzewitz, E. (2004). Prediction Markets. *Journal of Economic Perspectives*, 18(2), 107-126.
11. Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5-32.
12. Mangan, D., & Forkin, J. (2026, April 23). U.S. soldier arrested for Polymarket bets on Maduro capture that won \$400K, Justice Department says. *CNBC*.
13. Office of Rep. Ritchie Torres (2026). In response to suspicious Polymarket trade preceding Maduro operation, Rep. Torres introduces legislation to crack down on insider trading on prediction markets.